

Jacob Edwards, PhD

Data Scientist

[portfolio](#) · [github](#) · [linkedin](#) · [google scholar](#) | jacobedwards.jae@gmail.com | New York, NY

Data scientist with 13 years' building end-to-end pipelines on high-dimensional datasets in research environments. Specialized in large-scale biosensor and acoustic data (PhD, Columbia). I ship tools that cut workloads and turn quantitative results into clear visual stories. I bring disciplined reasoning and exceptional data storytelling skills to teams where the data is interesting and rigor matters.

SKILLS

Tools	Python, R, SQL, Excel, GitHub/git, Adobe Suite
Analysis	Statistical modeling, regression & classification, time-series analysis, geospatial analysis, NLP, machine learning, feature engineering, dimensionality reduction, model evaluation, data cleaning & visualization
Experimentation	Hypothesis testing, experimental design, sampling methods, effect size estimation, bootstrapping, simulation, metric design
Communication	Data storytelling, technical writing, oral presentation, cross-functional collaboration, teaching/mentoring

EXPERIENCE

Research Scientist

Columbia University · New York, NY · 2018 - 2025

High-volume sensor pipelines & quality automation

- Collected and analyzed 3.2 TB of high-resolution time-series data from 5.6k+ neurons across 20 subjects; built scalable ETL pipelines for cleaning, feature extraction, aggregation, and visualization.
- Automated data quality screening using an ensemble classifier model (SVM, RF, KDE) applied across 300+ recording sessions, reducing manual review time by 20%.
- Built custom interactive dashboards enabling non-technical collaborators to explore time-series data and turn complex quantitative results into audiovisual “movies” and sounds; adopted by 3 additional research teams.

Sequence modeling & behavioral analytics

- Collected and structured a dataset of 30k+ vocal events from 180 subjects; optimized an audio event segmentation pipeline to convert 20+ GB of raw recordings into analysis-ready tabular data.
- Built predictive NLP models to test sequence learning hypotheses, evaluating with a custom cost metric using fuzzy matching and bootstrapped effect size estimates.
- Engineered and systematically evaluated acoustic features using domain metrics and decomposition methods (PCA, wavelets); used cluster analysis on select features to identify class labels for NLP sequence models.
- Unified database and annotation infrastructure with a custom GUI, reducing work hours per project by 80% across 3 teams.

Statistics & Data Science Instructor

Columbia University · New York, NY · 2020 - 2024

Laboratory in Justice Data Science (2022, 2024)

[portfolio](#) ➔

- Designed and taught an original applied data science course covering R, data wrangling, visualization, machine learning, NLP, and network analysis using real-world policy datasets. Emphasized translating analytics into policy recommendations.
- Wrote 20 hours of original lectures and 7 hands-on assignments; received high student evaluations both semesters.
- One student capstone, a geospatial analysis of police use-of-force and domestic violence was published in the peer-reviewed journal *Violence Against Women*. Another capstone, a risk analysis of cybersecurity breaches related to AI and privacy rights, was published in the *Columbia Undergraduate Law Review*.

Statistics for Behavioral Scientists (2020, 2022, 2024)

- Led lab sections teaching statistical reasoning and R to 40+ students per semester for 3 semesters.

PROJECTS

SpikeJungle – Neural Data Quality Classifier

[portfolio](#) ➔

MATLAB · random forest · feature engineering · time-series

- Trained and deployed an ensemble classifier to automate data quality screening of electrophysiological time-series data, reducing manual review time by ~20%; trained on 400M+ labeled observations from 2 published datasets totaling 150+ GB.
- Designed for reuse: modular preprocessing and parameterized threshold allow adaptation to new recording configurations without retraining.

SylLabeler – Annotation & Database Management Tool

[portfolio](#) ➔

R · MATLAB · GUI · dimensionality reduction · clustering · database management

- Unified interface for segmenting, labeling, and managing 30k+ acoustic events across 180 subjects, reducing work hours per project by ~80%; adopted by 3 additional teams.

Ephys Explorer – Interactive Neural Population Visualizer

[portfolio ↗](#)

MATLAB · 3D visualization · interactive dashboard · time-series

- Real-time 3D visualization tool for exploring population-level neural responses across 5.6k+ neurons and both brain hemispheres from 5 species simultaneously.

EDUCATION

PhD, Psychology (Auditory Neuroscience)
MSc, Biology
BSc, Biology

Columbia University · 2025
Colorado State University · 2018
University of Tennessee, Knoxville · 2013

Presentations: Society for Neuroscience (2019, 2022) · Taste of Science NYC (2021) · Frontiers of Neuroscience, Barnard College (2021)

Peer-reviewed: 6 peer-reviewed publications · [Google Scholar ↗](#) · \$150k+ in competitive funding secured · NIH predoctoral grant (F31)